Irreversible cell injury
&
Tissue degeneration

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Objectives

- Describe irreversible cell injury
- List the different types of necrosis and their morphological changes.
- Describe cell death due to Apoptosis and its differences from necrosis
- List differences between dystrophic and metastatic calcification

Irreversible cell injury - Cell Death

- Necrosis
  - death of group of cells or tissues in a living person
- Apoptosis
  - programmed cell death
  - specific type of cell death involving single cell or a small group of cells in a tissue where the other cells are functioning normally

Other Mechanisms of Cell Death

- **Necroptosis**
  - Hybrid of necrosis + apoptosis
  - Morphologically like necrosis
  - Genetically regulated like apoptosis
- **Pyroptosis**
  - Inflammatory cell death
  - IL-1 release → fever
  - Occurs in infections
- **Ferroptosis**
  - Iron-dependent oxidative cell death
  - Due to lipid peroxidation when glutathione is depleted

Irreversible cell injury

- If the cause of cell injury persist the cell reaches the threshold where it is **no longer reversible.**

- Membrane damage
- Influx of extracellular Ca^{++} and release of intracellular Ca^{++}
- **Lysosomal swelling**
- **Mitochondrial vacuolization**
- **Pyknosis, karyolysis or karyorrhexis of nucleus**

Mechanisms of Irreversible cell injury

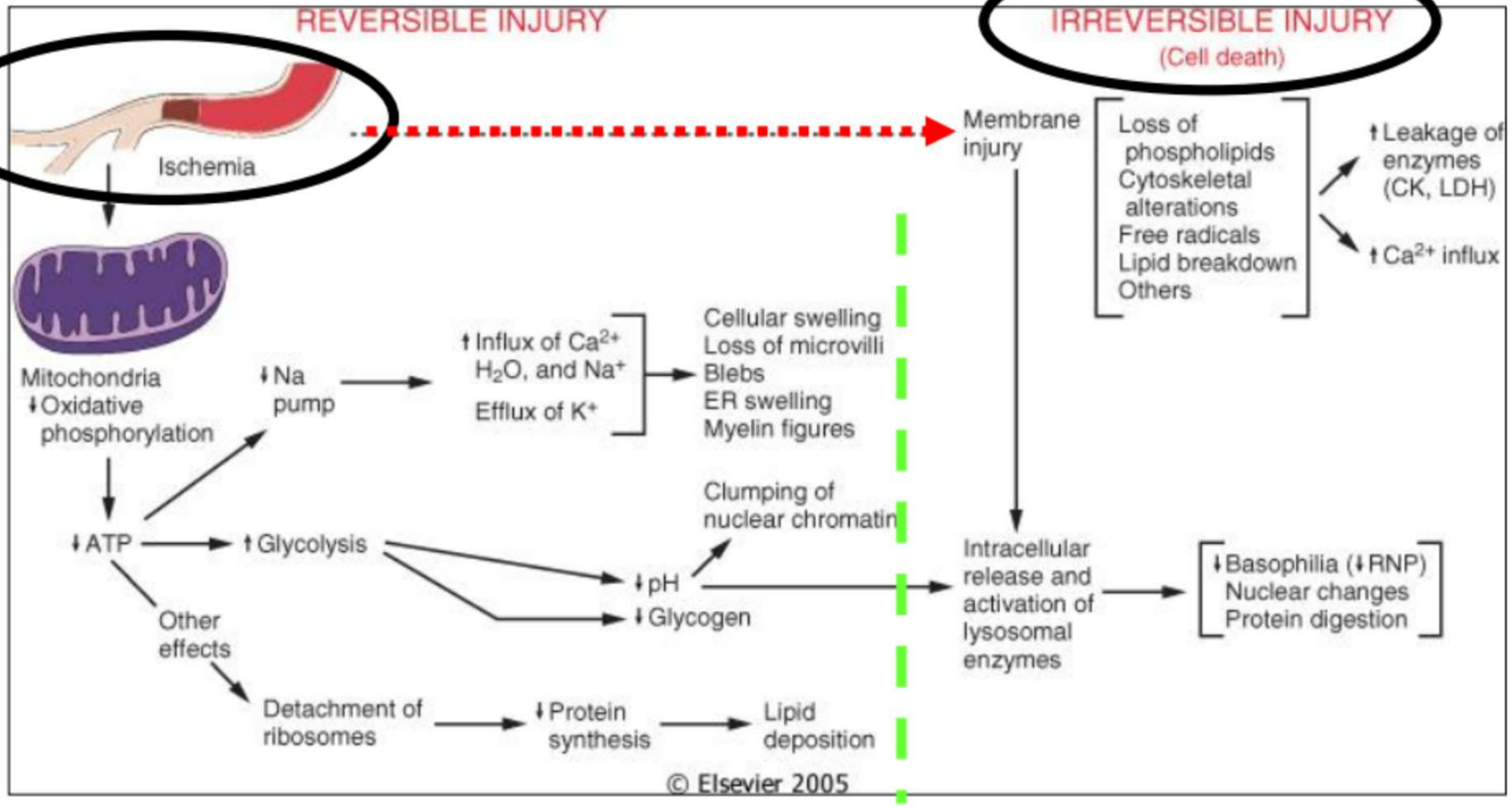
Two major mechanisms:

1. Mitochondrial dysfunction -

2. **Membrane function disorders** - the most central factor in the pathogenesis of irreversible cell injury

Mechanism of Irreversible cell injury-NECROSIS due to ischaemia

Sequence of Events in Ischaemic Cell Injury



Different types of Necrosis and Morphological changes

What is Necrosis?

- **death of group of cells or tissues in a living person**
- Necrosis refers to a sequence of morphologic changes that follow cell death

- **Types of necrosis:**

1. **Coagulative** necrosis (ischaemia)
2. **Liquefactive** necrosis (hydrolases)
3. **Fat necrosis** (enzymatic or non enzymatic)
4. **Caseous** necrosis
5. **Gangrenous** (ischaemia+bacterial liquefaction)
6. Fibrinoid necrosis (immune reactions)

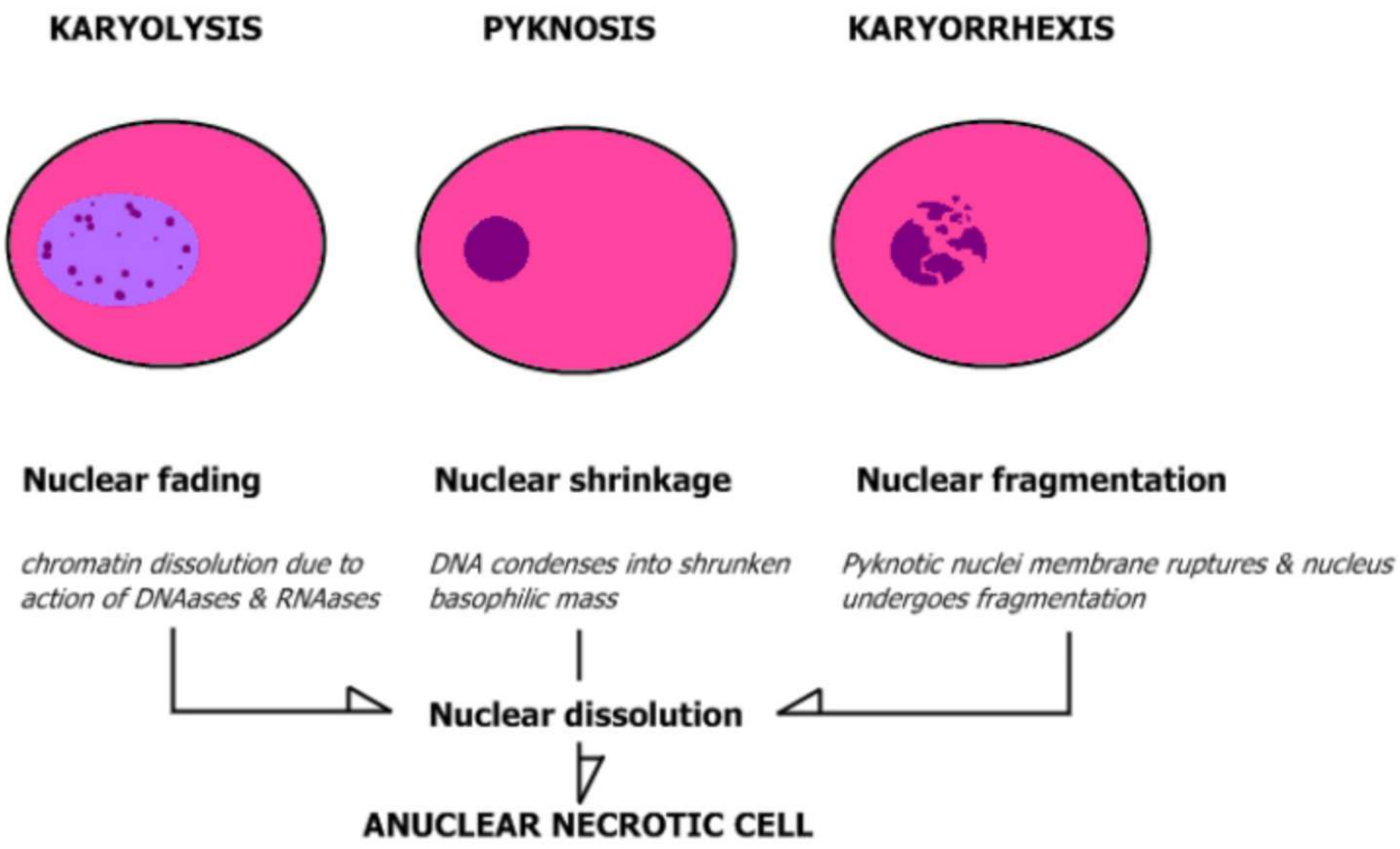
Morphologic appearance of necrosis

The result of **two processes:**

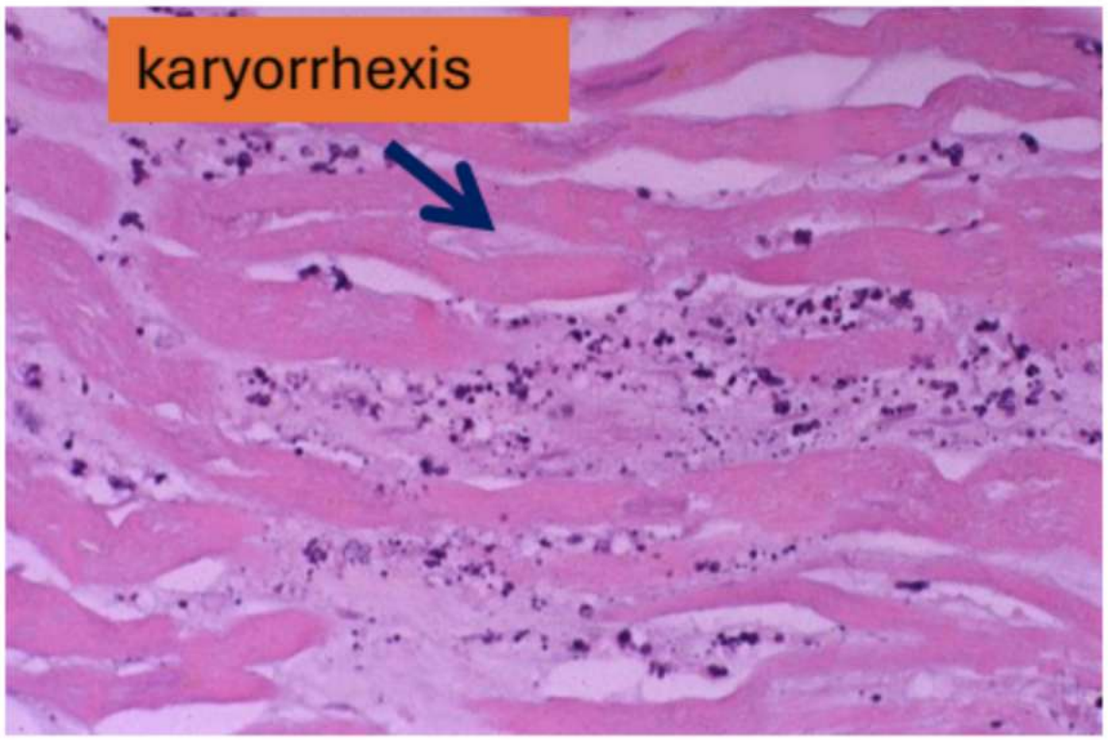
1. **Denaturation of proteins** coagulative
2. **Enzymatic digestion of the cell** liquefactive

Microscopically:

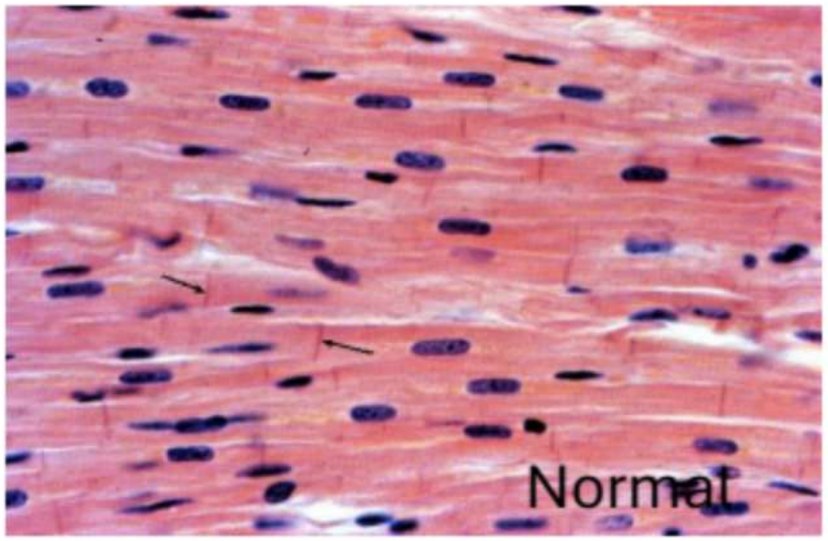
- Dead cells are more eosinophilic
- The cytoplasm becomes vacuolated, and the cell may be calcified.
- The nucleus becomes either pyknotic, undergoes karyorrhexis or karyolysis


KARYOLYSIS
PYKNOSIS
KARYORRHEXIS
Nuclear fading
chromatin dissolution due to action of DNAases & RNAases
Nuclear shrinkage
DNA condenses into shrunken basophilic mass
Nuclear fragmentation
Pyknotic nuclei membrane ruptures & nucleus undergoes fragmentation
Nuclear dissolution
ANUCLEAR NECROTIC CELL

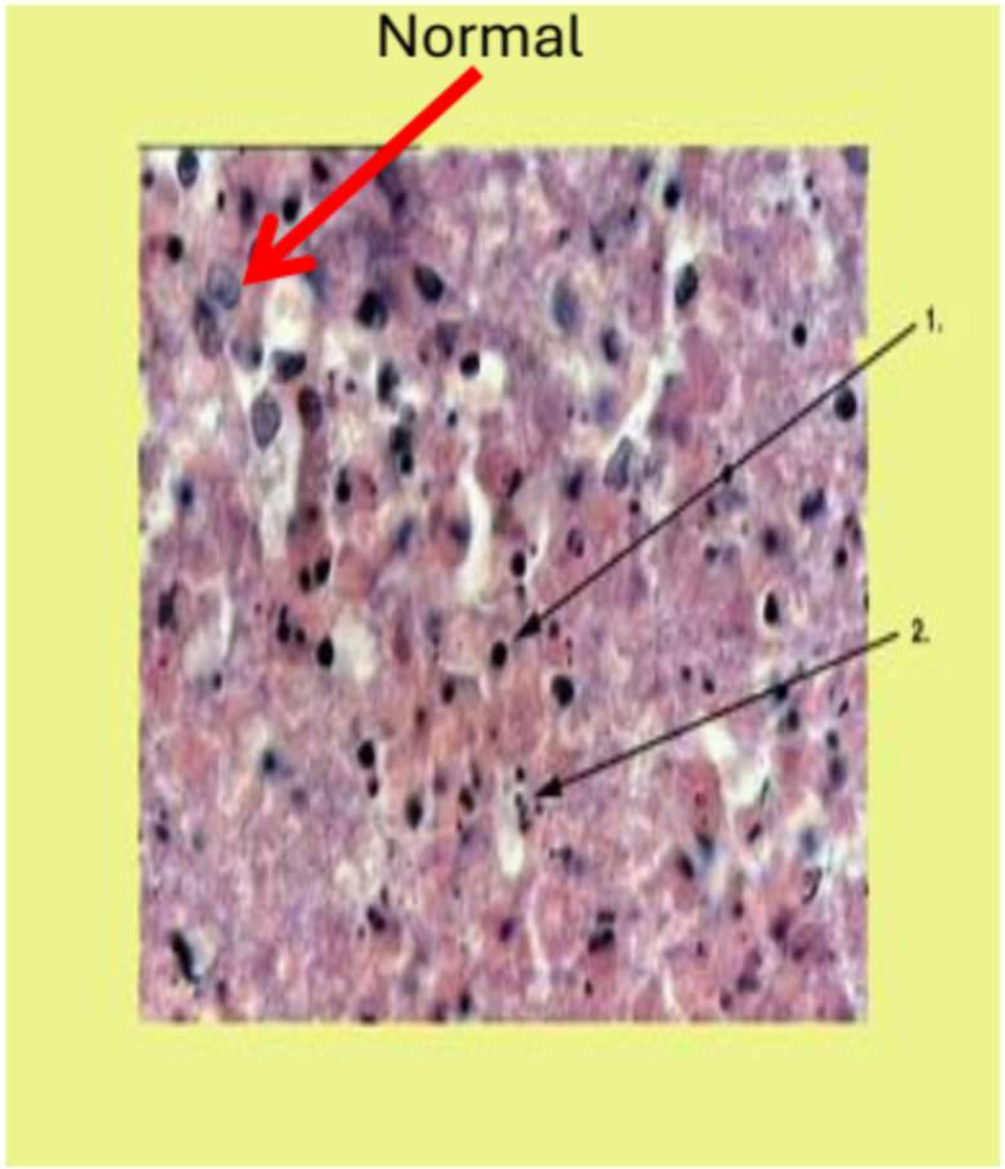
karyolysis



A



- Dead cells in the picture A are more eosinophilic due to loss of nuclear material etc
- Here, there is karyorrhexis or fragmentation of the nucleus. Also note the nuclear fading or karyolysis.
- Compare with the normal picture



- Arrow 1 shows a pyknotic nucleus or a shrunken nucleus. Compare that to the normal sized nucleus.
- Arrow 2 shows a fragmented nucleus or karyorrhexis

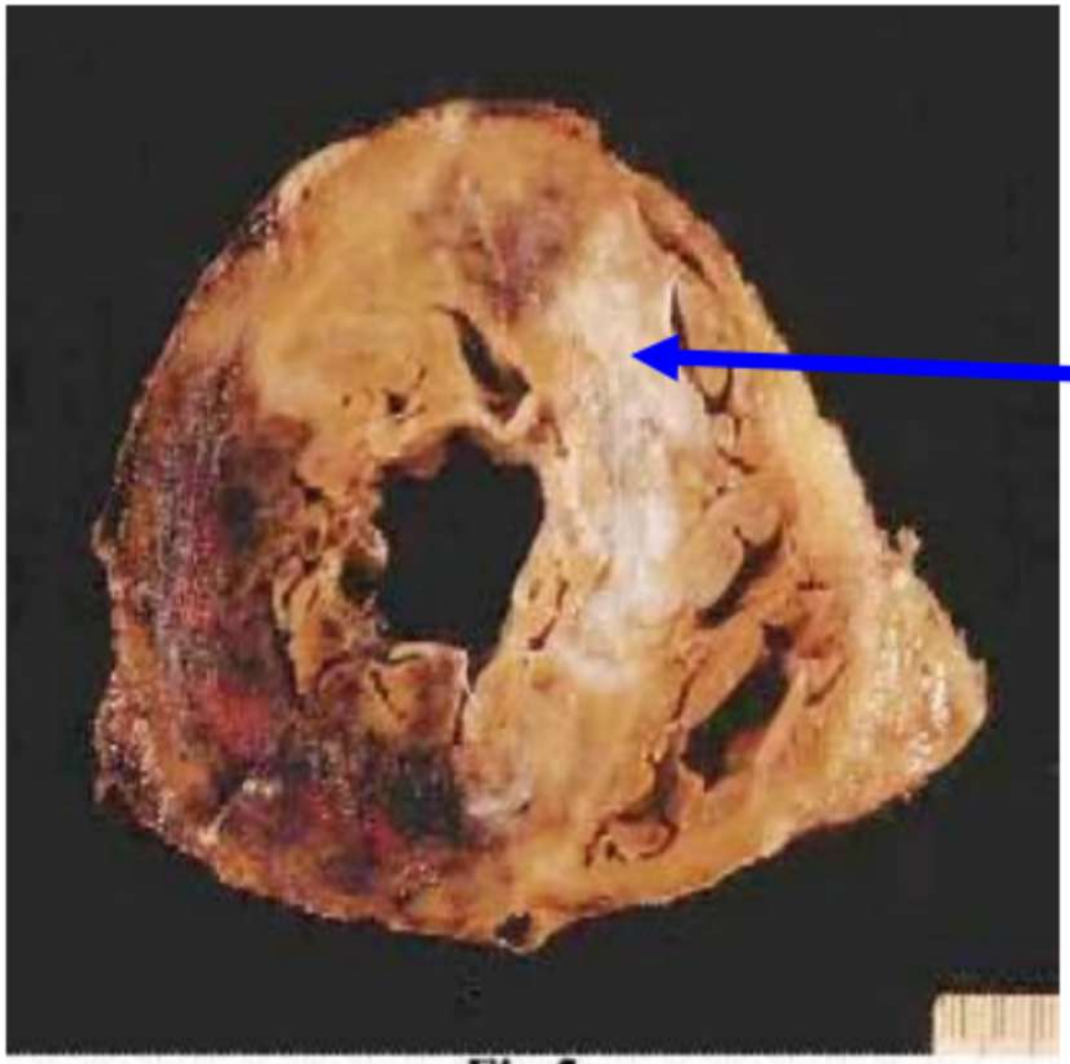
1. Coagulative necrosis

- Cause – ischaemic hypoxic injury,
  - **infarction** = tissue death (necrosis) caused by a local lack of oxygen, due to an obstruction of the tissue's blood supply
- Coagulation is caused by protein denaturation due to increased acidosis
- The general architecture is preserved because denaturation of enzymes
- Denatured enzyme are inactivated so it cannot break down protein
- Seen in most organ e.g. heart, muscle, spleen, gut

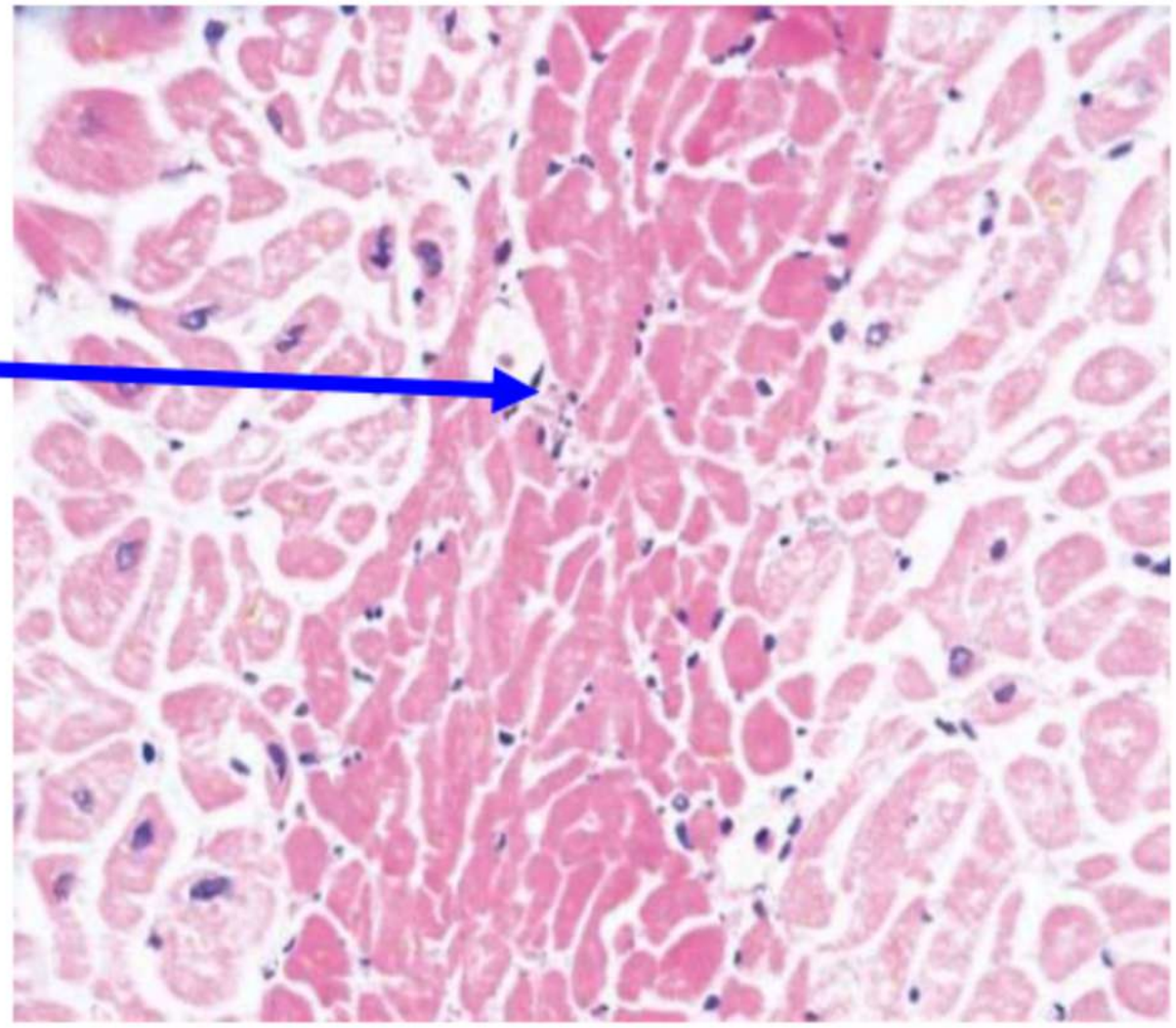




Coagulative necrosis



-Fig.2-



2. Liquefactive necrosis

- Seen in the brain and bacterial infection

Liquefactive necrosis in brain cells

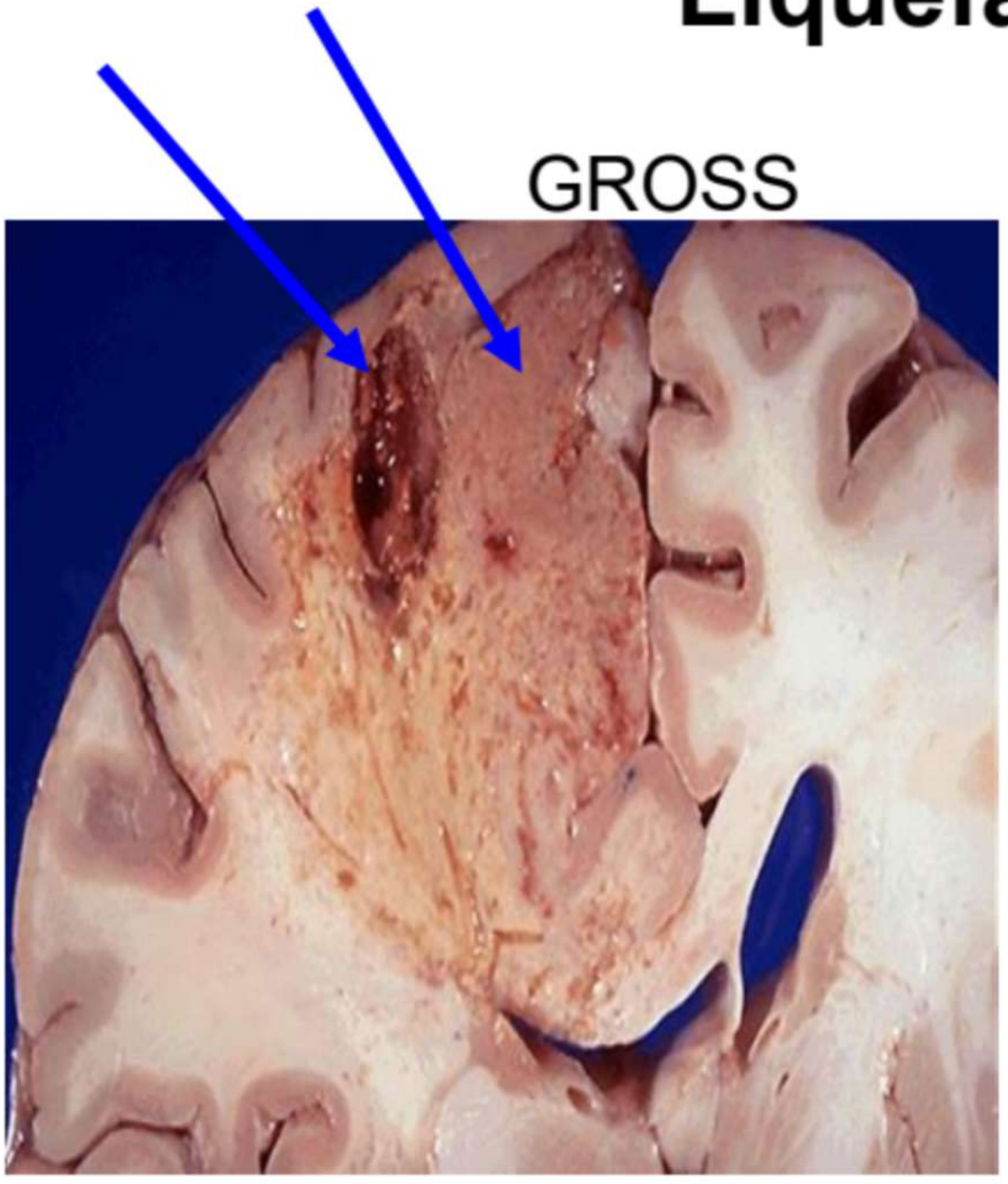
- Brain cells are rich in hydrolytic enzymes and lipids, and very little connective tissue
- Cells are digested by their own hydrolases
- Tissue becomes soft, liquefies, is walled off from healthy tissue, forming cysts

Liquefactive necrosis in bacterial infection

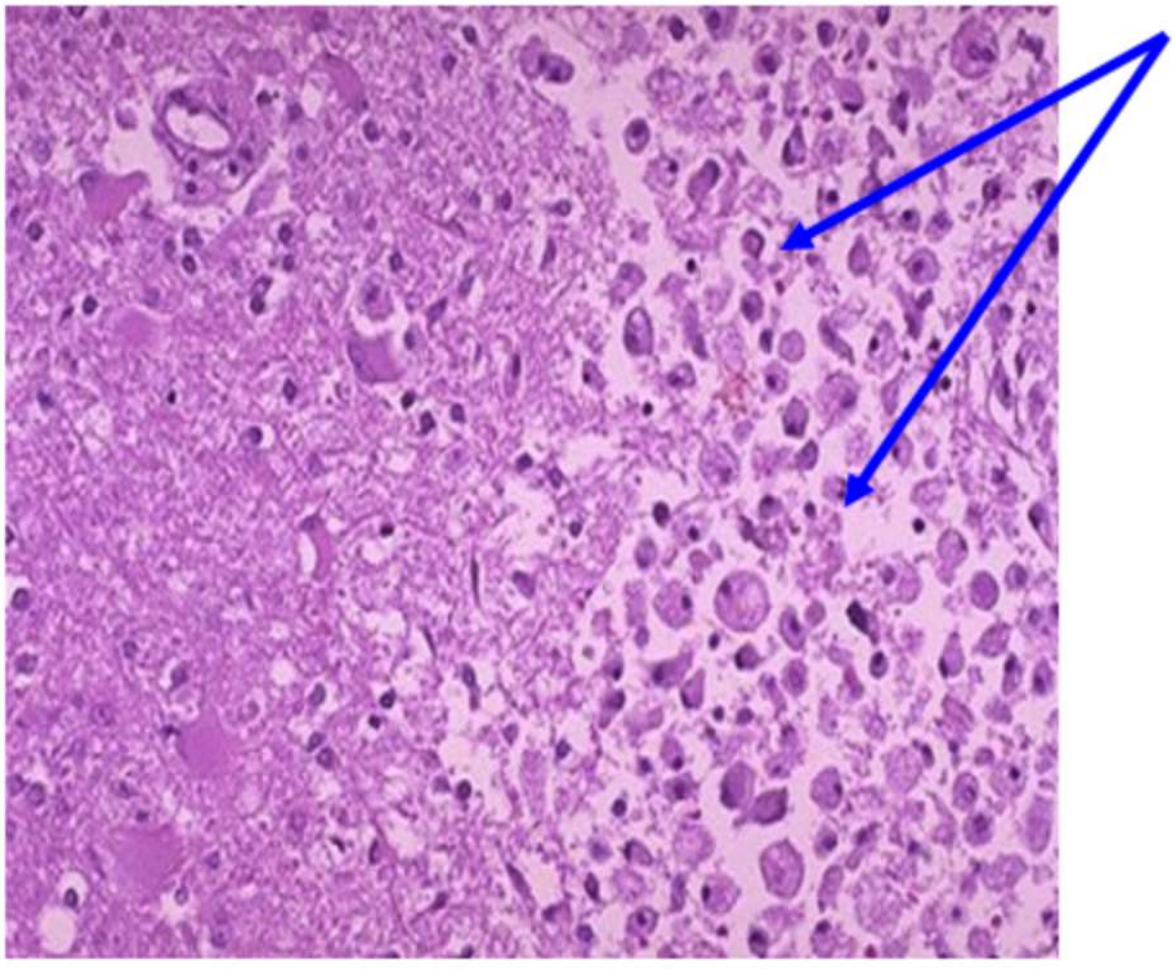
- Caused by bacterial infection by – *Staphylococcus, Streptococcus, Esch.coli*
- In bacterial infection, neutrophils release enzymes to destroy the bacteria. These enzymes also destroys the tissue causing liquefaction.

Liquefactive necrosis

GROSS



MICROSCOPIC



3. Fat Necrosis

- Seen in breast (traumatic fat necrosis), and pancreas (enzymatic fat necrosis)

Enzymatic fat necrosis

- Cellular dissolution caused by lipases
- Lipases breakdown triglycerides releasing fatty acids which combine with calcium, magnesium and sodium ions making soaps (saponification)
- Necrotic tissue appears opaque and chalky white

Traumatic fat necrosis

- After trauma to the breast, haemorrhage occurs
- This causes swelling of the breast tissue, with considerable tissue ischaemia and pressure necrosis, leading to necrosis of fat cells.

Fat Necrosis - Pancreas

Cellular injury to the pancreatic acini leads to release of enzymes which damage fat

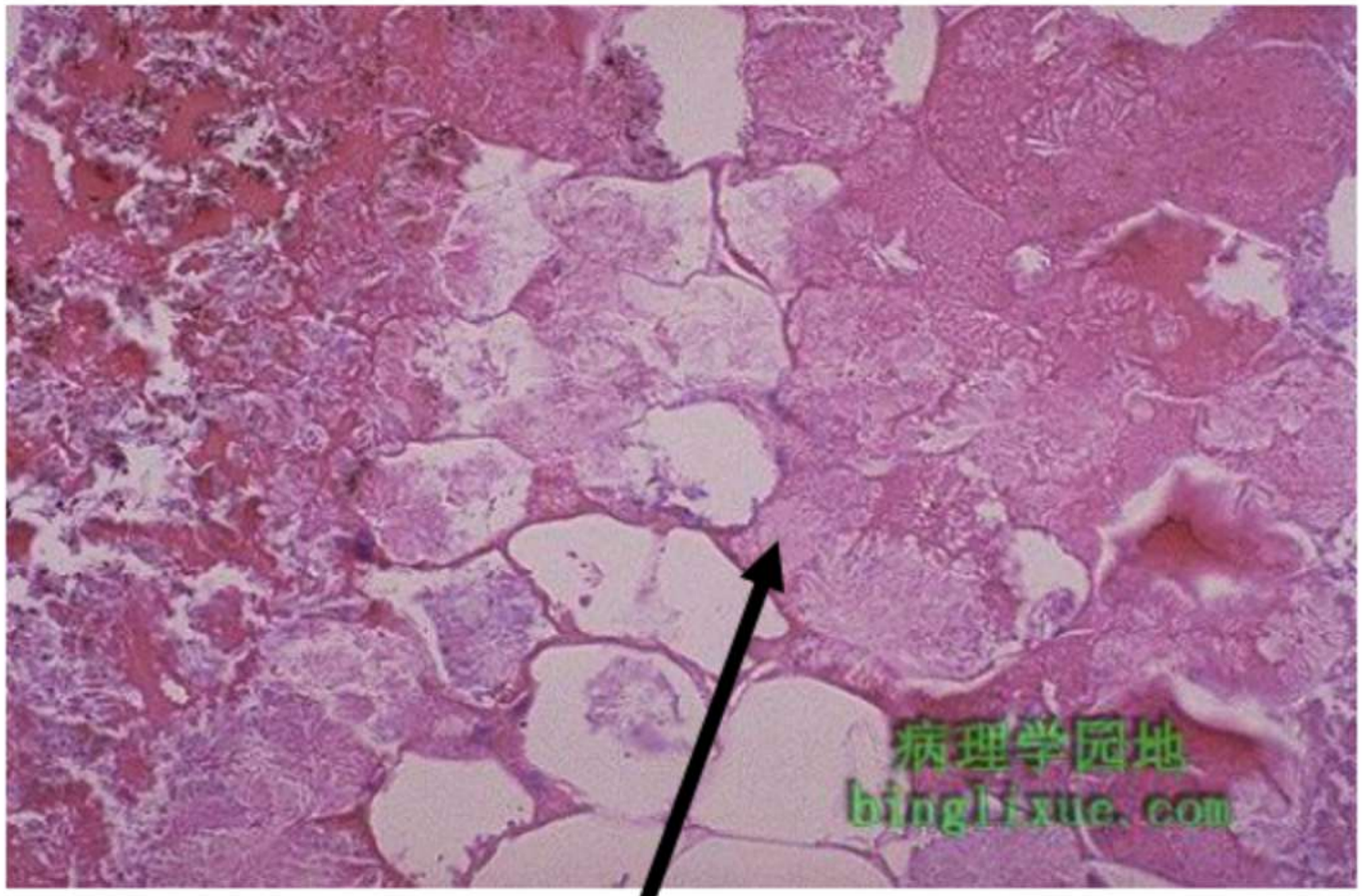
Gross



Fat Necrosis - Pancreas

Fat necrosis is seen here. The cellular outlines remains BUT the fat cells have lost their nuclei and their cytoplasm has become a pink amorphous mass of necrotic material.

Microscopic:



Blurred cell boundary with loss of cell nuclei - Pancreas

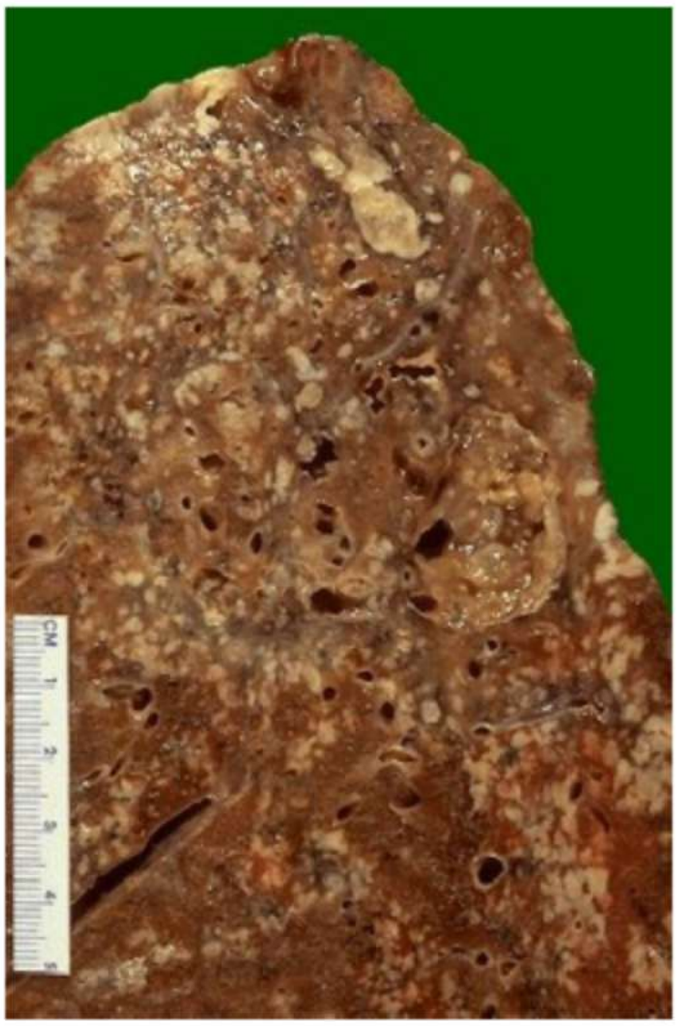
4. Caseous Necrosis TB

- It is characteristic of tuberculous infection, can involve any organ
- Mycobacterium tuberculosis **cell wall** contains a complex wax (peptidoglycolipids) which causes toxic effect
- Dead cells disintegrated but **not completely** digested by hydrolases
- On gross it has cheese like consistency, soft gray friable
- Microscopically there is granuloma or tubercle with central area of caseous necrosis



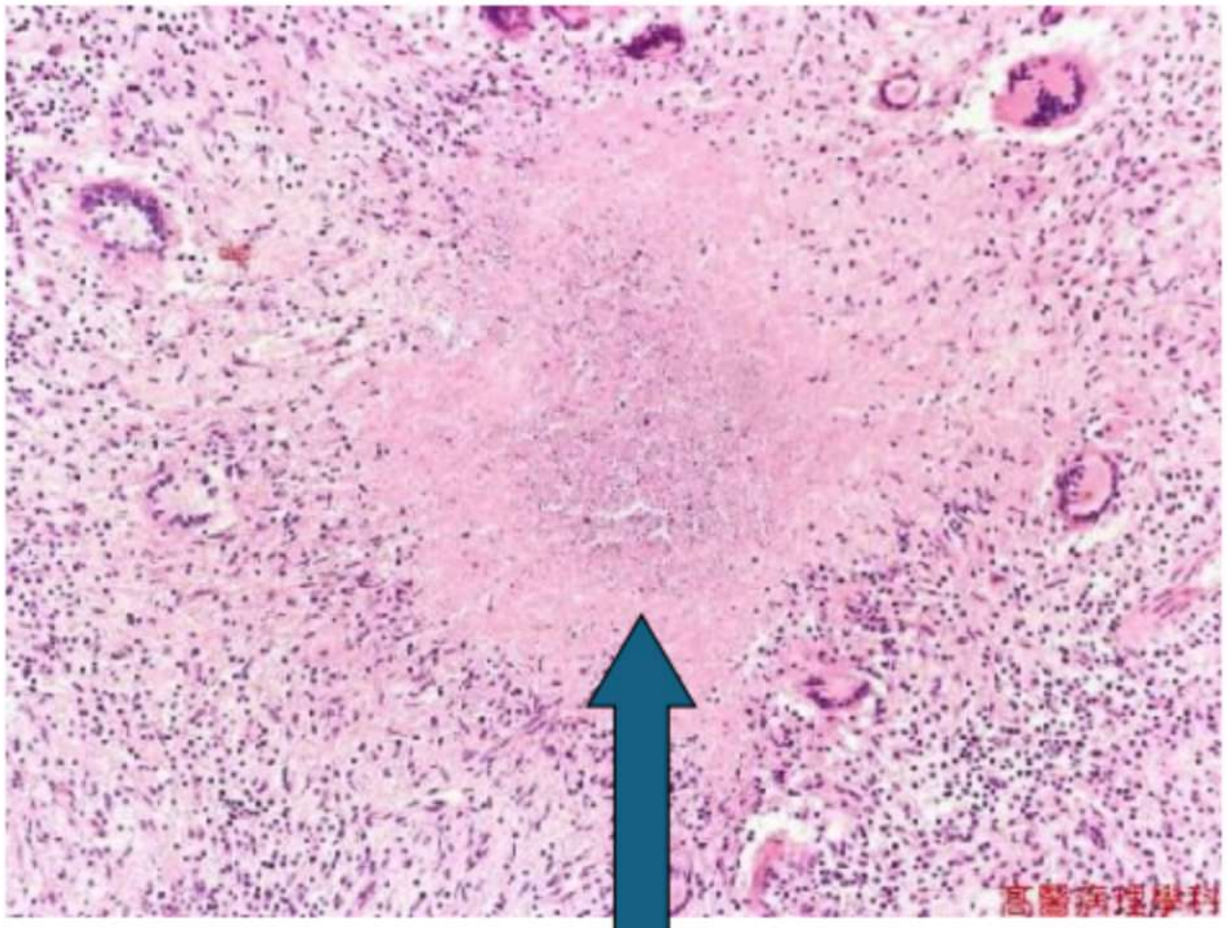
Caseous Necrosis

Gross



Tuberculosis - Lung

Microscopic



Caseous necrosis

5. Gangrenous Necrosis

- Mostly seen in lower **extremities and bowel**
- Death of tissue due to hypoxia**.**
- Dry gangrene –coagulative necrosis without liquefaction - skin becomes dry, wrinkled and dark brown (dusky) in colour
- Wet gangrene - When complicated by infection and invasion of neutrophils causing liquefactive necrosis.

Gangrenous Necrosis

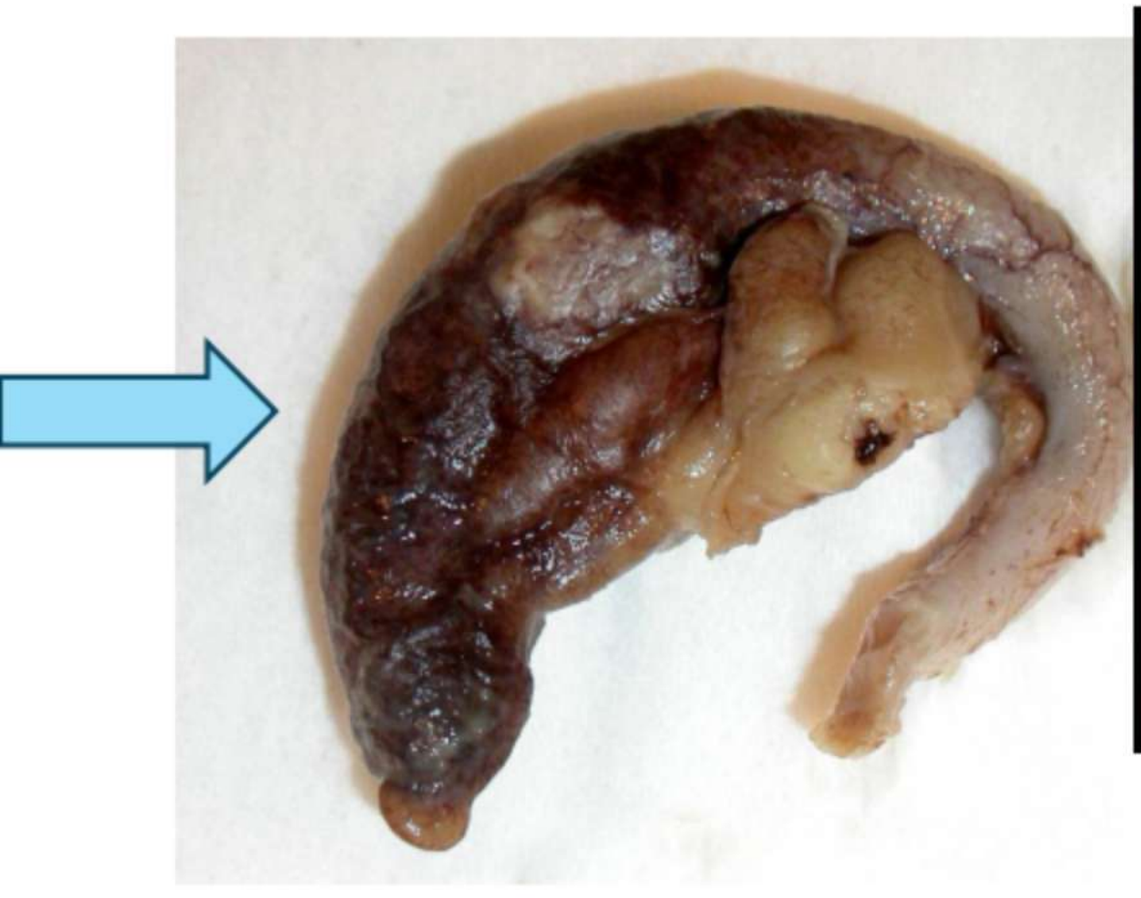


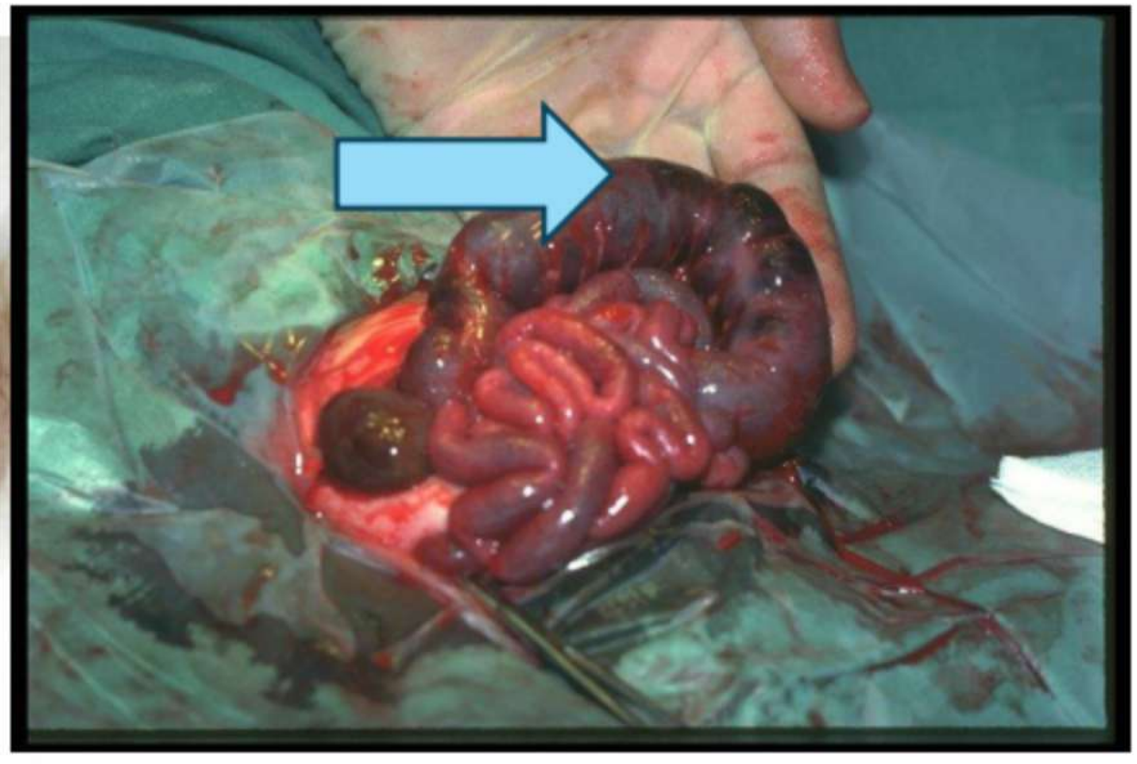
This is gangrene, or necrosis in a part of foot. The toes were involved in a frostbite injury. This is an example of "dry" gangrene in which there is mainly coagulative necrosis from the injury. **There is NO pus**



"wet" gangrene = there is liquefactive component from bacterial infection on top of coagulative necrosis from ischemia. This patient had diabetes mellitus. Usual bacteria = Staphylococcus or Streptococcus sp. **There is pus**

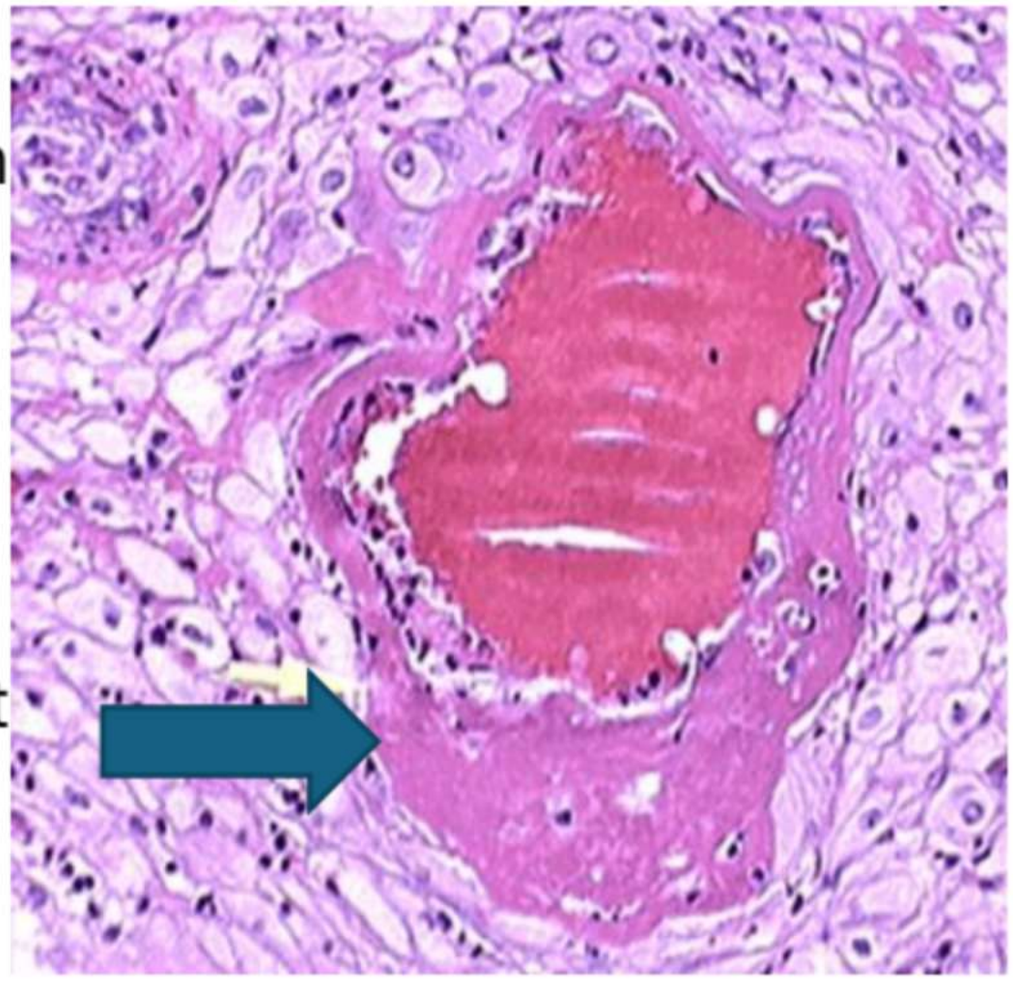
Gangrenous bowel





FIBRINOID Necrosis

- Necrosis associated with immune complex deposition involving blood vessels.
- Seen when complexes of antigens and antibodies are deposited in the walls of arteries
- These complexes will activat complement system and neutrophilic activity.
- Immune complex deposits caused the bright pink and amorphous appearance



What is the Difference between Reversible vs Irreversible Cell Injury?

Reversible injury

- Decreased ATP levels
- Ion imbalance
- Decreased pH
- Hydropic Swelling
- Fatty change

Irreversible injury

- Severe membrane damage
- Lysosomal rupture
- Extensive DNA damage
- (Karyorrhexis, karyolysis, pyknosis)

Systemic manifestation of Cellular Injury

Systemic manifestation of Cellular Injury

Manifestation	Cause
Fever	Endogenous pyrogens (IL-1,TNF-α, Prostaglandins, acute inflammatory response)
Tachycardia	Increased oxidative metabolism due to fever
Leucocytosis	Due to infection
Pain	Release of bradykinins, Prostaglandin E2 obstruction, pressure
Presence of cellular enzymes in blood	Release from tissue cells: LDH,CK,ALT,AST,ALP,Amylase& aldolase

Clinical Assessment of Cell Injury

- **Clinical observation**
- **Enzyme release:**
  - Cardiac enzymes
    (Myoglobin, CkMB, LDH, Troponin)
  - Liver enzymes
    (AST, ALT, Alkaline Phosphatase)
  - Pancreas (Lipase, Amylase)
- **Biopsy:** Liver, Lung, GI Tract, Skin etc

APOPTOSIS

Apoptosis

- is also known as

  “Programmed cell death”

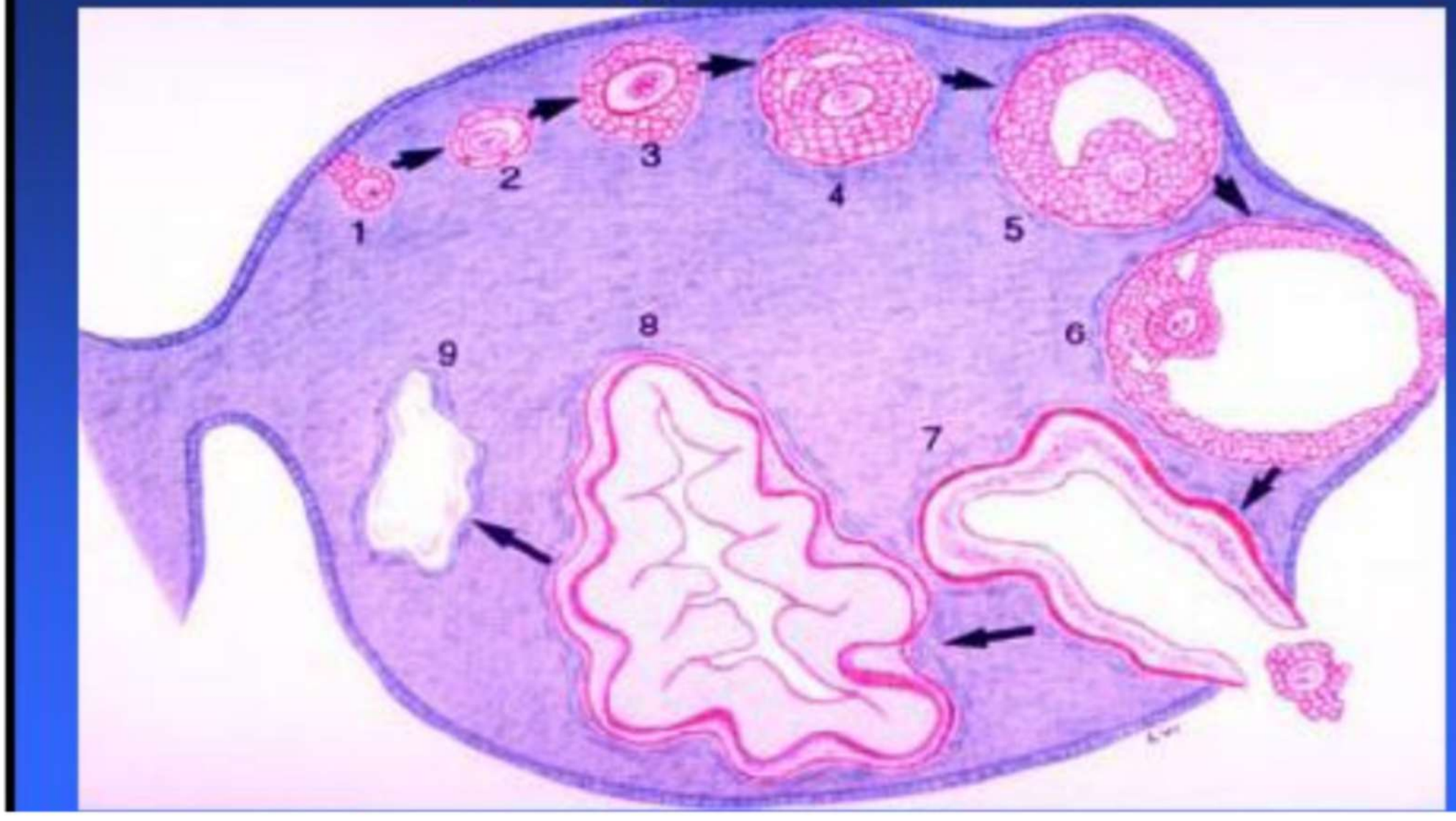
- This is an important mechanism for removal of cells: cells with irreparable DNA damage (from free radical, viruses, cytotoxic immune mechanism), protecting against malignant transformation
- It can occur in normal tissue for regulating number of cells or cell removal during embryogenesis

Physiological Apoptosis

Examples:

- Embryogenesis – formation of digit from limb buds, formation of lumen in gut
- Menstrual cycle – endometrial cell loss
- Ovulation
- Breast – after cessation of lactation
- Immune cell development – deletion of immune cells (T) that may react with the body's own tissues

Physiological apoptosis



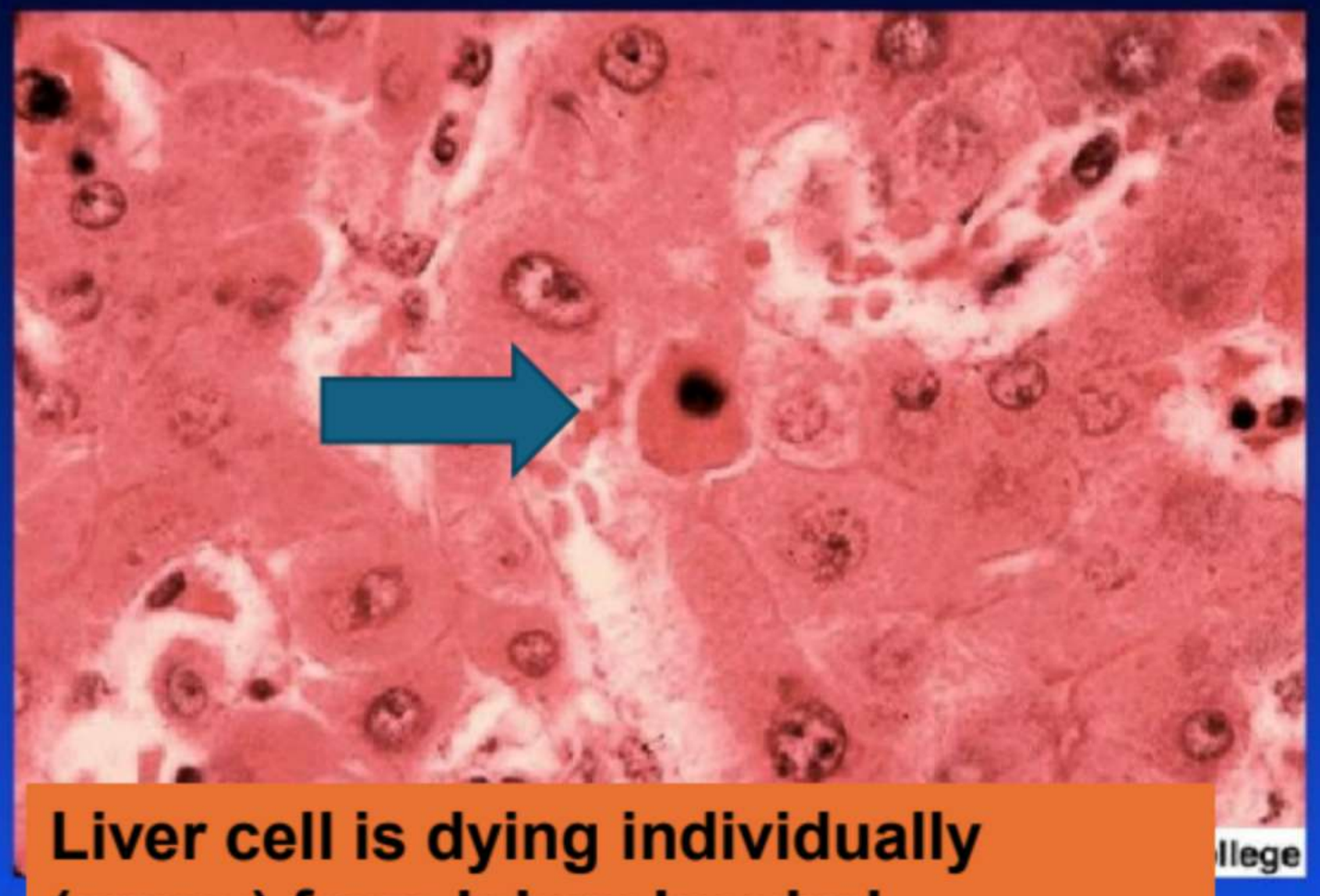
Ovulation

Pathological Apoptosis

Examples:

- **Tumours** – Apoptosis is the major cancer killing mechanism. When tumour forms,balance between apoptosis and cell proliferation is disturbed
- **Viral infection**– e.g. viral hepatitis, infected individual hepatocyte can be seen in apoptotic form
- **AIDS** – loss of CD4 T-lymphocytes is by apoptosis

Apoptosis in Viral Hepatitis

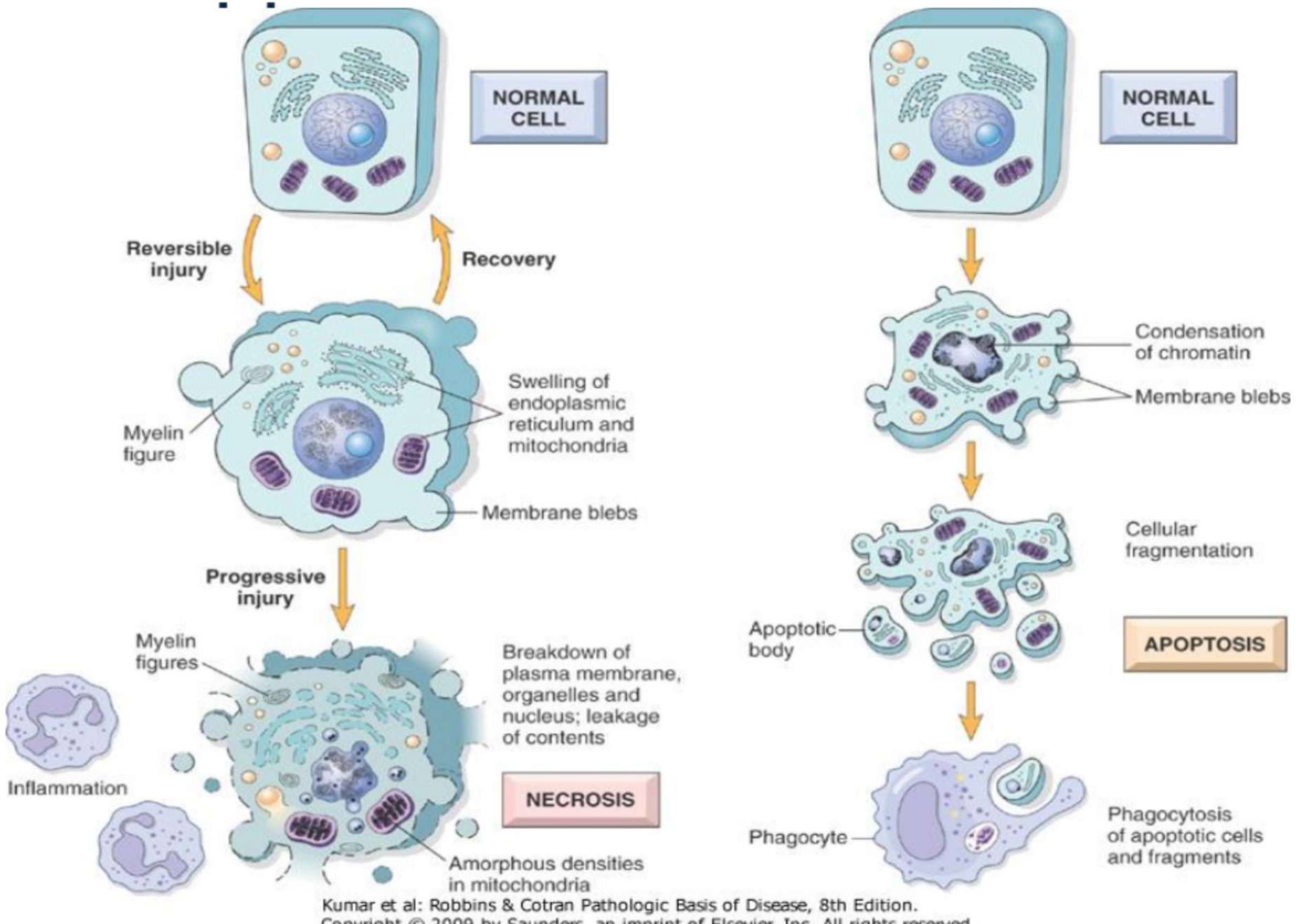


Liver cell is dying individually (arrow) from injury by viral hepatitis. The cell is pink and with pyknotic nuclei. Note the absence of inflammatory cell.

Apoptosis vs. Necrosis

Apoptosis	Necrosis
Physiologic	ATP depletion
Viral	Membrane damage
Immune	Free radicals
Cell shrinkage with water loss	Cell swelling
Membranes and organelles intact	Membrane blebs
Cell and nuclear fragmentation	Organelle breakdown
Internucleosomal DNA fragmentation	“Ghost cell” residual
Minutes	Hours
Apoptotic bodies (phagocytosed)	Inflammation

Reversible and Irreversible Cell



Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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Summary

- With continuing damage, the injury becomes irreversible and cell cannot recover
- Irreversibly injured cell undergoes morphologic changes recognised as cell death
- There are two types of cell death: Necrosis and Apoptosis, which differ in morphology, mechanism, role in disease and physiology
- Necrosis is always a pathologic process, with different morphologic types

Dystrophic vs Metastatic Calcification

- **Dystrophic Calcification**
- Occurs in **dead or damaged tissues**
- **Normal serum calcium**
- No systemic calcium metabolism disorder
- Mitochondria are often first to calcify

- **Metastatic Calcification**
- Occurs in **normal tissues**
- Due to **hypercalcemia**
- Causes: hyperparathyroidism, bone destruction, vitamin D intoxication

Dystrophic calcification

- ❑ Calcification in dead tissues

  - Necrosis
  - Infarcts
  - Thrombi
  - Hematoma
  - Dead parasites

- ❑ Calcification in degenerative tissues

  - Dense old scars
  - Atheromas
  - Stroma of tumours (leiomyoma)
  - Monckeberg’s sclerosis

Metastatic calcification

- May occur in normal tissues whenever there is hypercalcemia.

- FOUR primary causes of hypercalcemia:
  - hyperparathyroidism
  - resorption of bone e.g. metastasis to the bone
  - vitamin D - related disorders e.g. Vit D intoxication
  - renal failure, causing phosphate retention and secondary hyperparathyroidism

- Metastatic calcification may occur widely throughout the body but principally affects the interstitial tissues of the gastric mucosa, kidneys, lungs, systemic arteries, and pulmonary veins.
- All of these tissues excrete acid and therefore have an internal alkaline compartment that predisposes them to metastatic calcification.

Metastatic Calcification

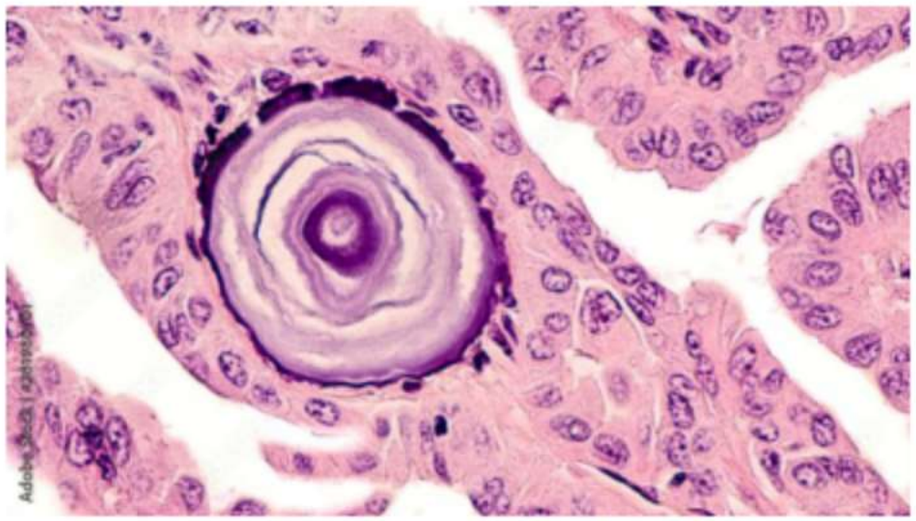
- Occurs in **normal tissues** due to **hypercalcemia**.
- **Causes of hypercalcemia (4):**
  - Hyperparathyroidism
  - Bone destruction (e.g. metastasis)
  - Vitamin D intoxication
  - Renal failure → secondary hyperparathyroidism
- **Common sites:**
  - Gastric mucosa
  - Kidneys (especially tubular basement membranes)
  - Lungs
  - Systemic arteries
  - Pulmonary veins
- **Why these sites?**
- All these organs have regions where **acid is actively secreted → leaving the interstitial environment alkaline**. Calcium salts (especially calcium phosphate) are **less soluble in alkaline conditions**. When the interstitial fluid is alkaline, **calcium and phosphate ions precipitate more readily**, forming deposits in tissues.

Morphology

- **Macroscopic**: vary in size, fine, white granules or clumps, often felt as gritty deposits.

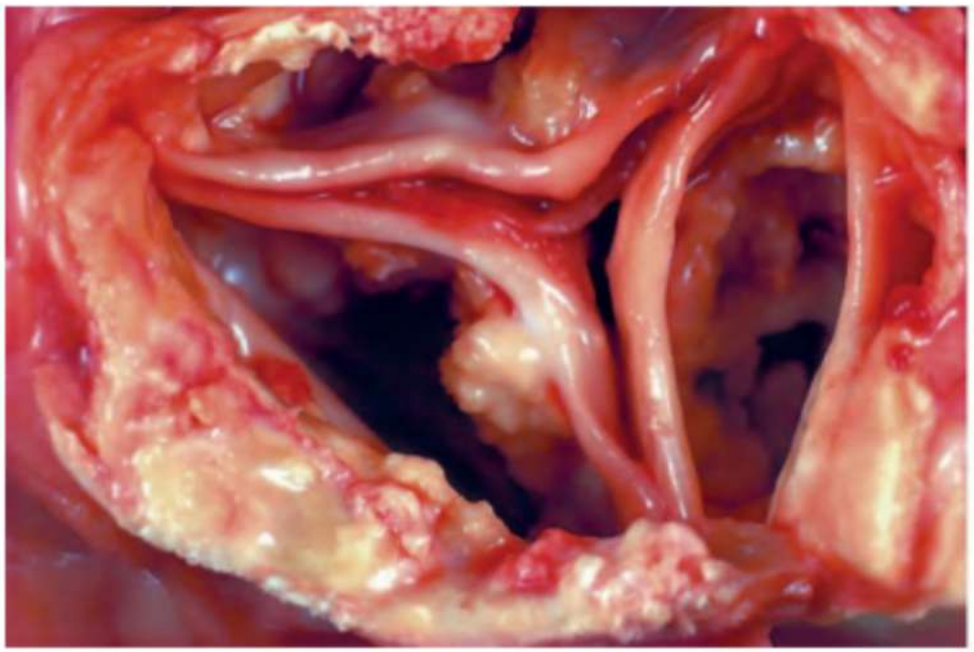
- **Microscopic**:
  - basophilic, amorphous granular / clumped appearance.
  - intracellular, extracellular or both locations in H & E stain.

* Psammoma bodies: concentric laminations of calcium deposition (seen in papillary thyroid carcinoma)



Pathological Calcification

- Dystrophic calcification of the aortic valve



Gross: The semilunar cusps are thickened and fibrotic, and behind each cusp are irregular masses of piled-up dystrophic calcification.

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